

PORTFOLIO

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Nuryabilla Utami, S.T.

Fresh Graduate 2026



Geodetic Engineering

Universitas Diponegoro

INTRODUCTION



 Cirebon, Indonesia

EDUCATION

Universitas Diponegoro 2022-2026

S1 - Geodetic Engineering

CGPA 3.57/4.00

ABOUT ME

Fresh graduate Bachelor's of Geodetic Engineering Universitas Diponegoro (2026) with internship experience in **surveying, mapping, GIS, and remote sensing**. Proficient in spatial data processing using **ArcGIS, QGIS, AutoCAD, Google Earth Engine, ENVI, SNAP, and Agisoft Metashape**, as well as data management and automation using **Python (Geospatial), SQL, and PostgreSQL**. Experienced in operating field surveying instruments such as **Total Station, Waterpass (Automatic Level), and GNSS**. Detail oriented and ready to contribute professionally in the engineer and geoscience field.

SOCIAL MEDIA AND CONTACT

 [linkedin.com/in/nuryabillautami](https://www.linkedin.com/in/nuryabillautami)

 github.com/nuryuwii-cue

 nuryabillau@gmail.com

 082262904885

Curriculum Vitae (CV)

 visit me on:
Website Portfolio

MY EXPERTISE ON

GEODETIC ENGINEERING

CORE EXPERTISE

SURVEYING & MAPPING

Utilizing ground surveying instruments (Total Station, Waterpass, GNSS) and CAD software (AutoCAD/Civil 3D) to perform **stakeout**, **generate contours**, **create cross & long sections**, **calculate cut & fill volumes**, and **deliver accurate as-built drawings**.

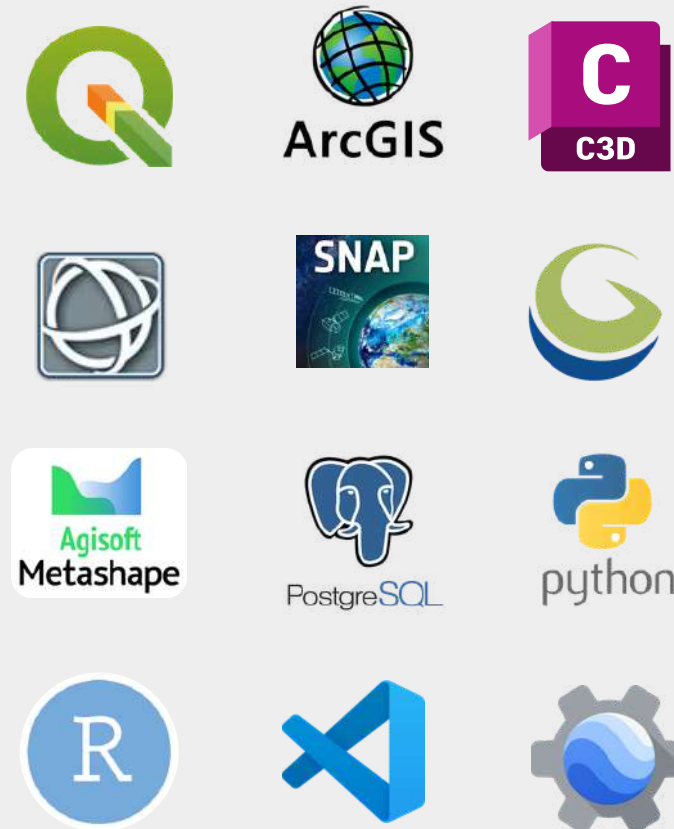
GIS & REMOTE SENSING

Utilizing ArcGIS, QGIS, SQL, ENVI, and Python to process spatial data, integrate active & passive remote sensing data processing (**SAR, NDVI, Landuse**) and **create thematic maps** to generate insights and support machine learning models.

PHOTOGRAMMETRY

Planning UAV flight paths and processing aerial & close-range photogrammetry data using Agisoft Metashape, Global Mapper, and Trimble software to generate **high-resolution 3D models**, **Digital Elevation Models (DEM)**, and **Digital Surface Models (DSM)**.

TOOLS & SOFTWARE



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nuryabillau@gmail.com



082262904885



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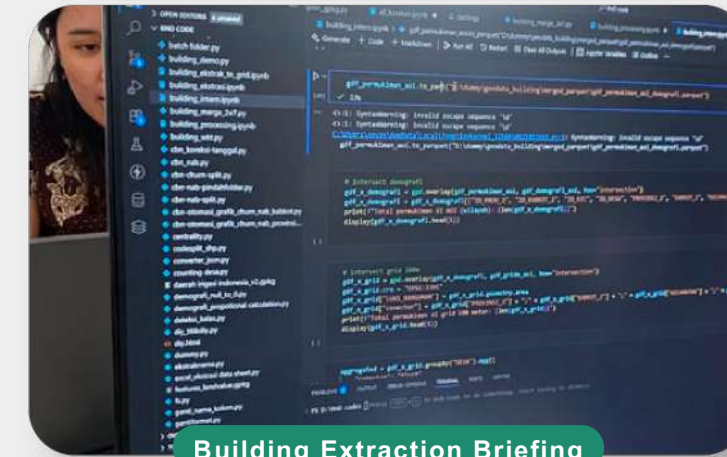
INTERNSHIP EXPERIENCES

GIS ANALYST INTERN

PT MULTI AREAL PLANING INDONESIA (MAPID)

June - August 2025

- Surveys potential POIs across 9 cities (Kedungsepur & Sarbagita) to update spatial catalog data.
- Processed building footprint downscaling (25 cities) and LST data using Python.
- Completed an individual project titled "Spatial Analysis of Potential Employment Absorption Areas by District in Tasikmalaya City, 2024" using QGIS.



Building Extraction Briefing



Weekly Progress Report



Intern Final Presentation



Intern Final Presentation

SURVEYOR INTERN

ATR BPN KOTA CIREBON

January - February 2025

- Land parcel surveys and prepared survey drawings/plans using AutoCAD for land certificate documentation.
- Performed verification and examination of land certificate documents and legal records for 22 sub-district in Cirebon.



Land Survey



BPN Media Support



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nuryabillau@gmail.com



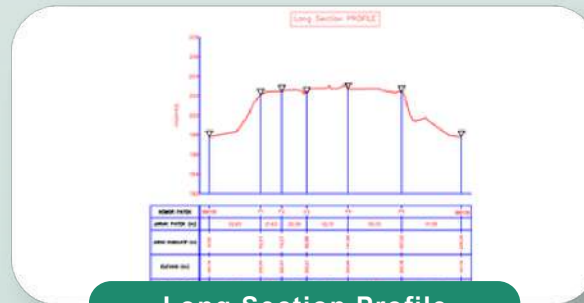
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ACADEMY EXPERIENCES

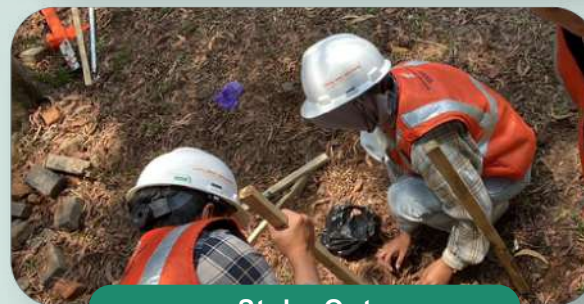
PRACTICUM: TOPOGRAPHIC SURVEYING



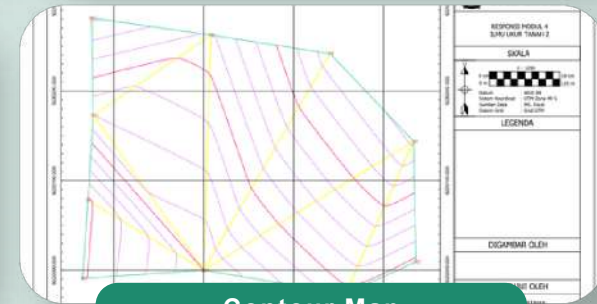
Long Section Profile



Topographic Surveying



Stake Out



Contour Map



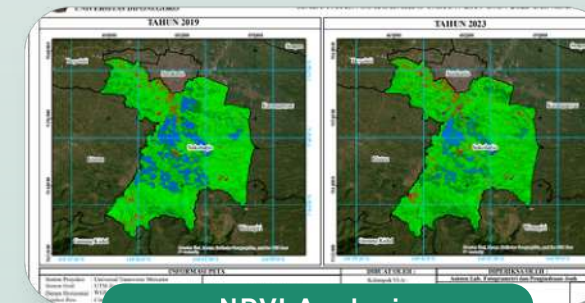
As Built Drawing



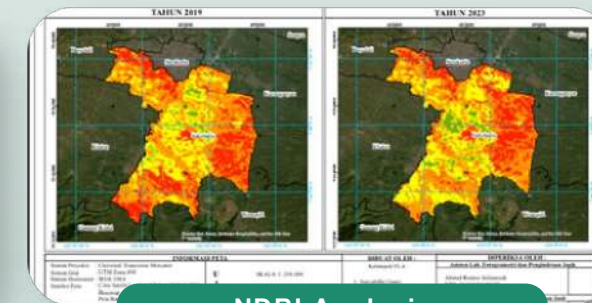
Drafting Survey Data

- Executed terrestrial and GNSS surveys using Total Station, Theodolite, Waterpass, leveling rod, and measuring tape for contour, long, and cross-section mapping.
- Calculated cut & fill volumes, performed stake-outs, and drafted as-built drawings using AutoCAD Civil 3D.

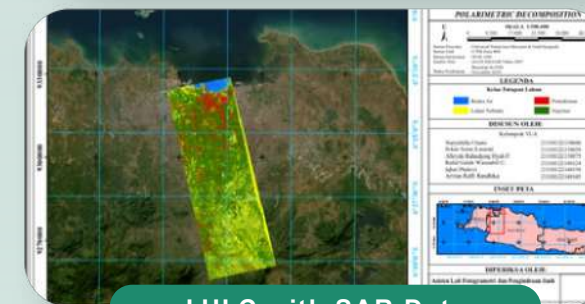
PRACTICUM: REMOTE SENSING



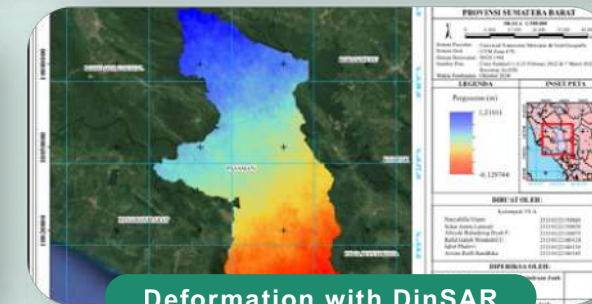
NDVI Analysis



NDBI Analysis



LULC with SAR Data



Deformation with DinSAR



LULC with Segmentation



Group Discussion

- Mapped land use from Landsat 8 using supervised and unsupervised machine learning methods, and extracted spectral indices (NDVI, NDBI, NDWI, LST).
- Processed Sentinel-1 SAR and ALOS PALSAR data using DInSAR for deformation analysis and land use using Polarimetric Decomposition techniques.



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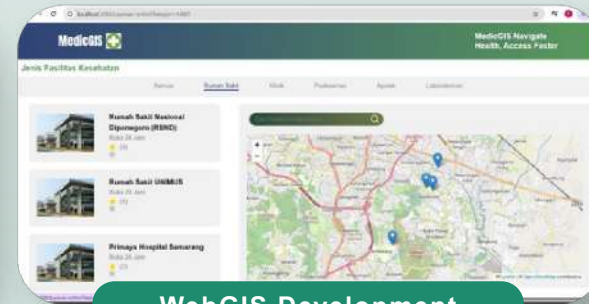
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ACADEMY EXPERIENCES

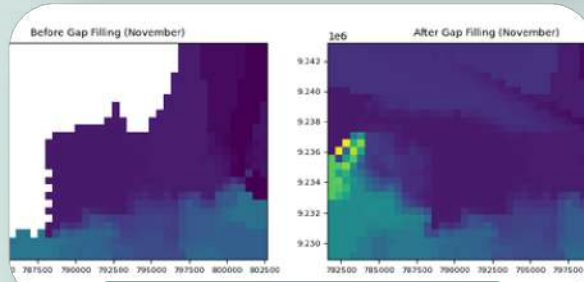
PRACTICUM: GIS & DBMS



Relational DB on PostgreSQL



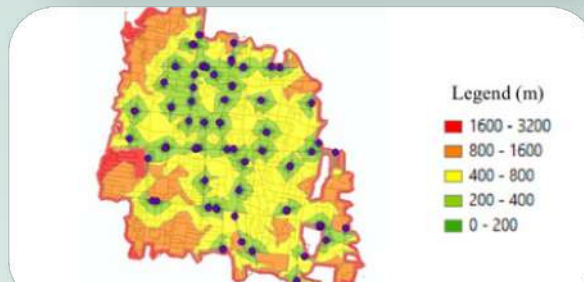
WebGIS Development



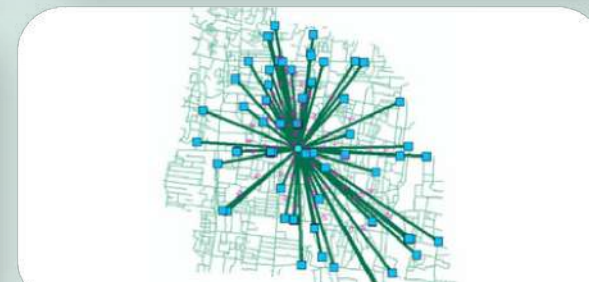
Machine Learning Modeling



Geodatabase Visualization



Service Area Analysis



OD Cost Matrix Analysis

- Designed ERD models for PostgreSQL database management and developed WebGIS applications for Health Facilities in Tembalang.
- Applied Machine Learning modeling for geospatial data, create Service Area, OD Cost Matrix, and Closest Facility analysis.

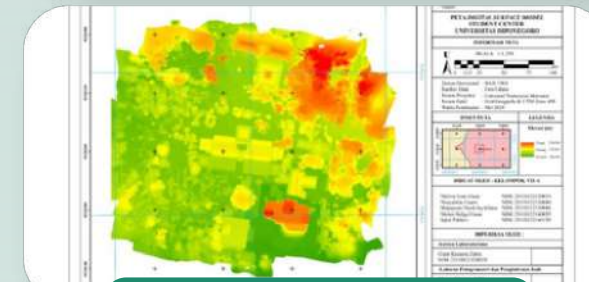
PRACTICUM: PHOTOGRAMMETRY



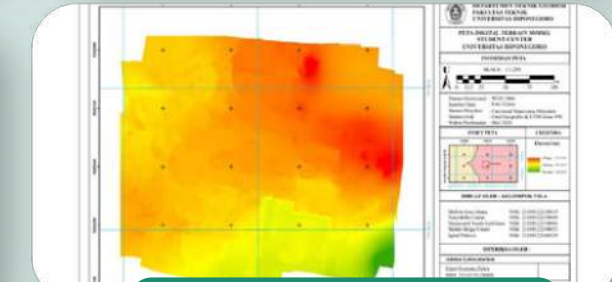
Fathoming Drone Operation



Flight Path Planning



DSM Map



DTM Map



Orthophoto Map



3D Model Data Acquisition

- Designed flight paths for aerial photo acquisition and generated stereo photo pairs.
- Created ortho-mosaics and DEM/DSM/DTM models.
- Reconstructed 3D object models using Close Range Photogrammetry techniques.



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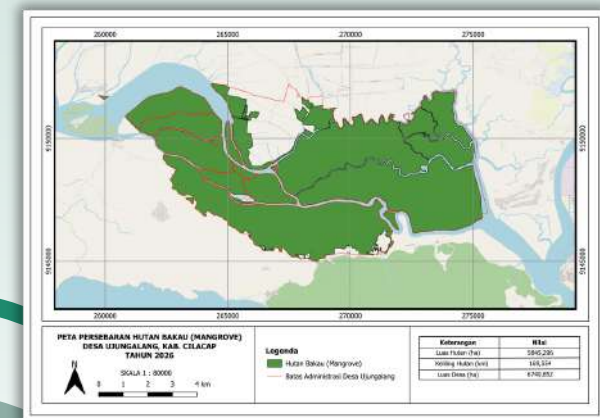
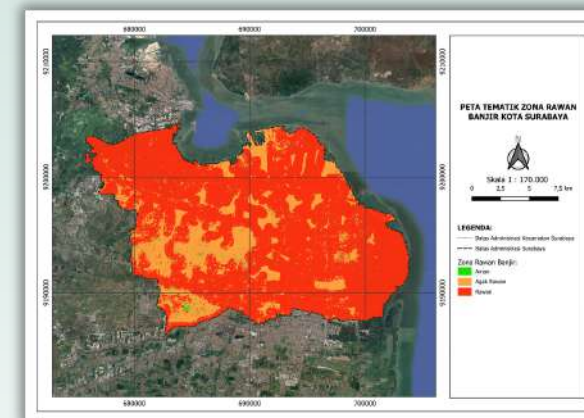
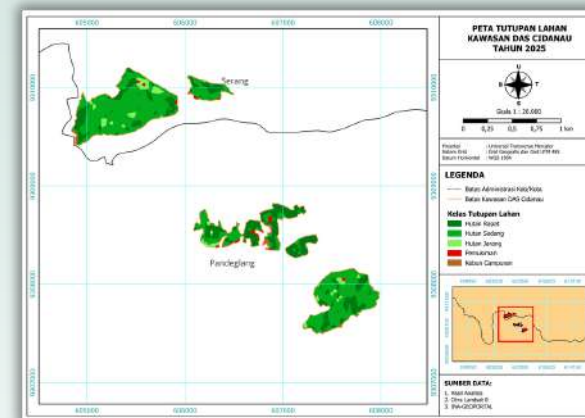
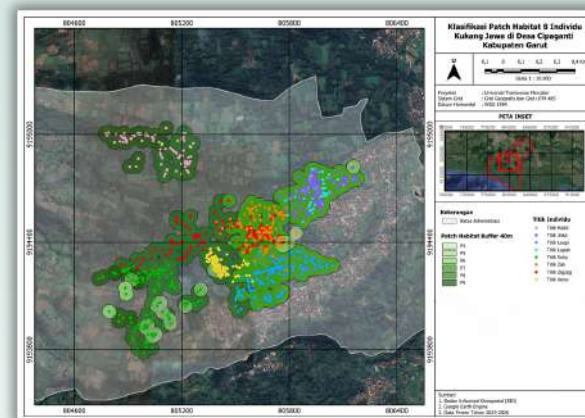
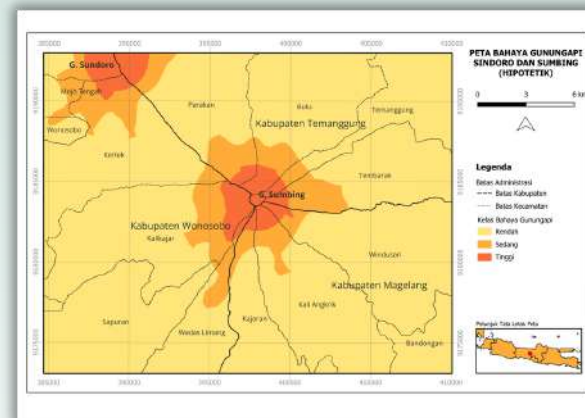
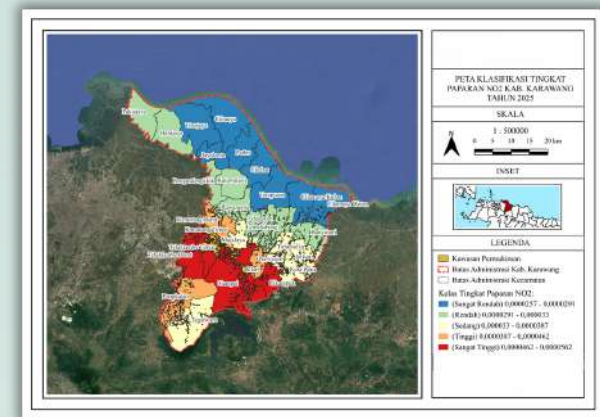
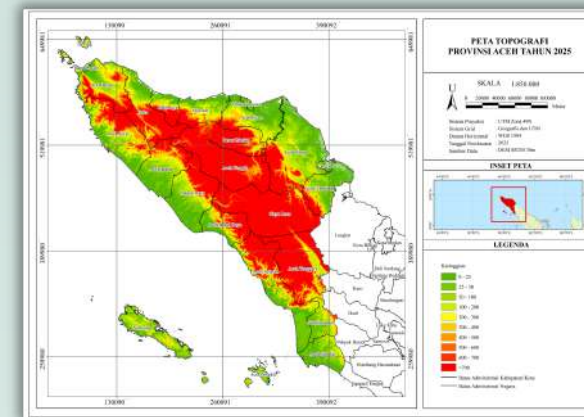
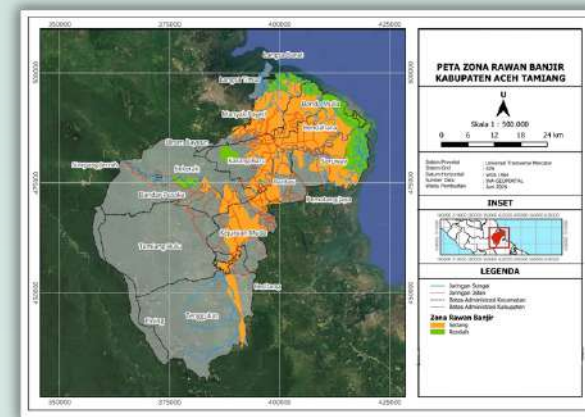
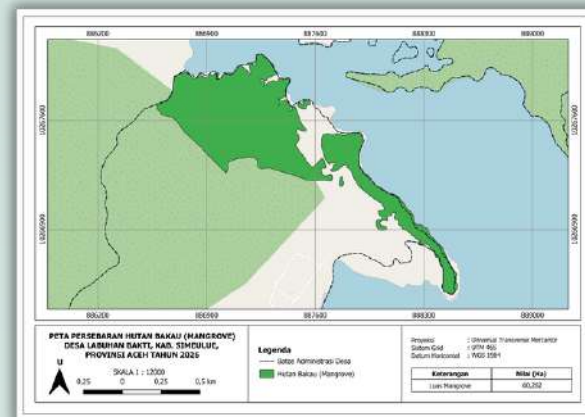
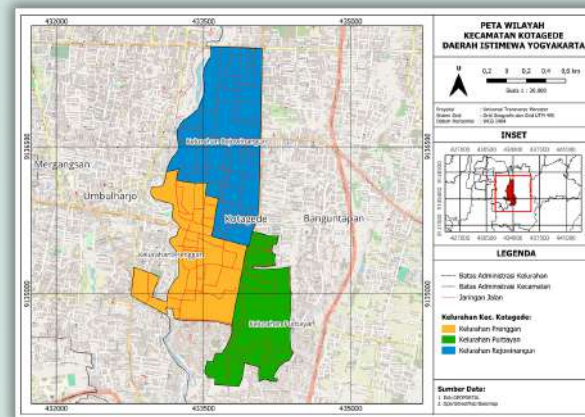
PROFESSIONAL EXPERIENCES

GIS SPECIALIST - FREELANCE

May 2026 - Present

- Completed 25+ geospatial projects, including thematic map production and technical consulting in geospatial fields for clients from academic and private sectors.

VIEW MORE



[linkedin.com/in/nuryabillautami](https://www.linkedin.com/in/nuryabillautami)



nuryabillau@gmail.com



082262904885



github.com/nuryuwii-cue

ACADEMY PROJECT

FINAL PROJECT RESEARCH (2026)

Geostatistical Modeling of NO₂ Concentrations and Its Impact on Acute Respiratory Infections (ARI) Prevalence in Mid-2025 in Semarang City



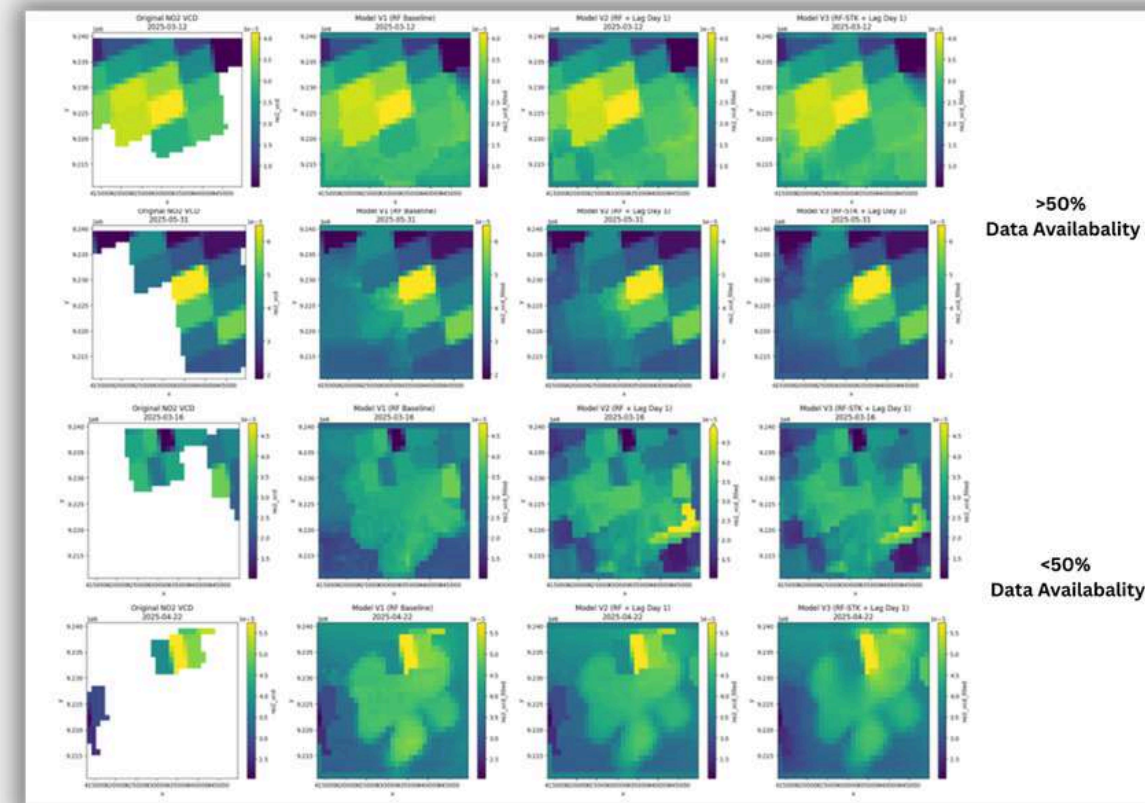
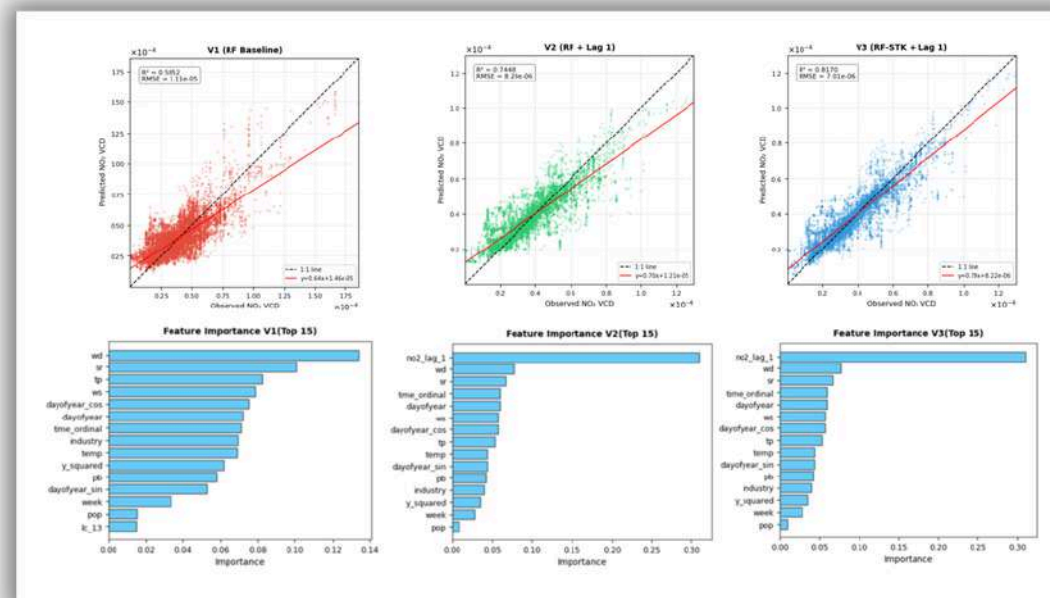
OPEN PROJECT

BACKGROUND OF RESEARCH

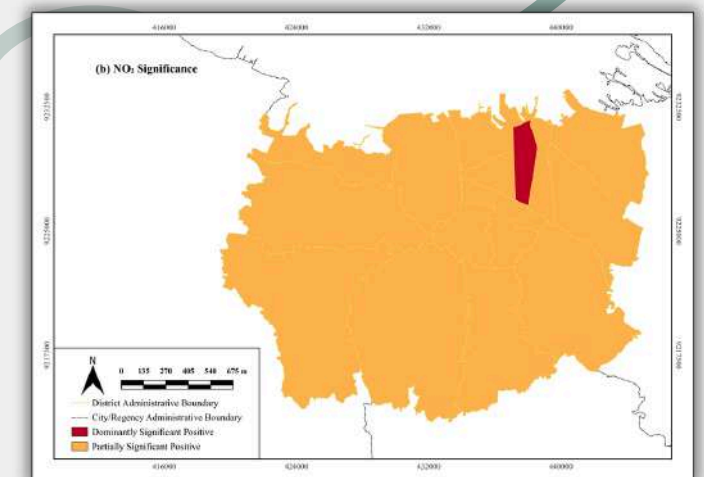
- Semarang recorded **AQI 147** and **154k+ ARI cases** in mid-2025.
- Assesses carcinogenic NO₂ concentration influence on high local ARI prevalence.
- Employs daily **Sentinel-5P satellite monitoring** while **addressing data gaps** caused by cloud cover and QA filter limits (≥ 0.75) using machine learning model (RF0, RFLagDay1, RFSTK).

RESULTS & ANALYSIS

RF-STK (V3) yielded **highest accuracy** ($R^2 = 0.80$), driven by NO₂ lag-1, wind direction, and solar radiation.



- **>50% Data:** V3 achieved best spatial coherence via Kriging, outperforming V1 (coarse) and V2 (lag-1 feature).
- **<50% Data:** Reliability dropped for all models as predictions relied more on predictors than satellite data.



- GTWR superior to OLS (R^2 : 0.681; Moran's I: 0.467).
- Universal **positive** NO₂-ARI impact, strongest in **East Semarang** (traffic/industry) and peaking in **dry season** (Jan & Feb-Mar).



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nuryabillau@gmail.com



082262904885



github.com/nuryuwii-cue

ACADEMY PROJECT

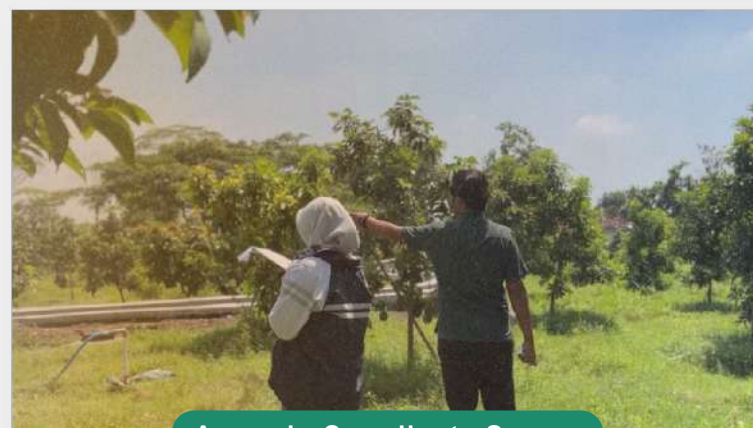
KKN INDIVIDUAL PROJECT (2025)

PROJECT 1

Distribution Map of Avocado Ketapang Tree Varieties of BUMDes.

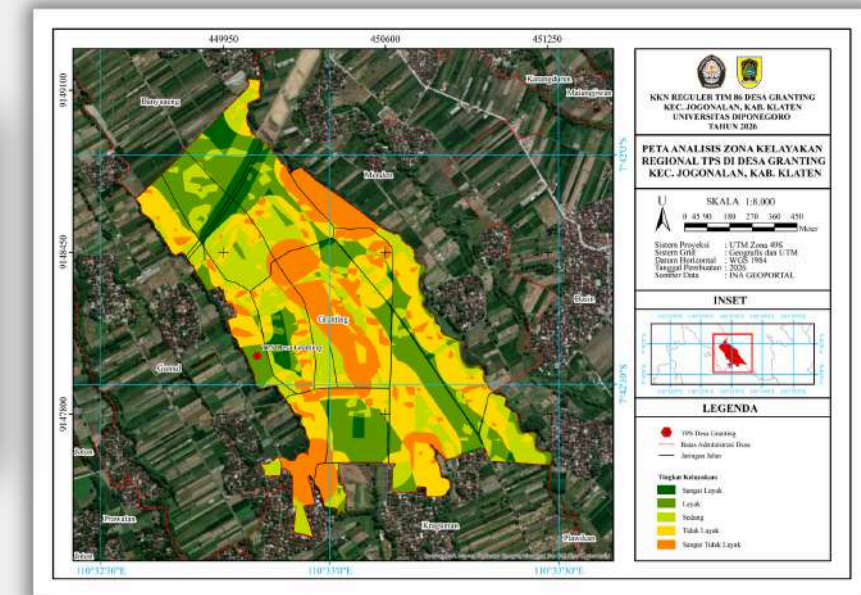


This avocado variety distribution map, derived from the spatial database, serves as a data-driven tool for BUMDes managers to plan harvests, monitor tree health, and develop targeted marketing and orchard management strategies.



PROJECT 2

Development of a Regional Feasibility Zone Analysis Map for Waste Disposal Sites (TPS) in Granting Village, Jogonalan District, Klaten Regency.



This map helps Granting Village make data-driven decisions to select and manage safe, sustainable TPS 3R sites. It supports environmental and social risk mitigation, improves spatial planning, and guides future village waste management and development programs.



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nuryabillau@gmail.com



082262904885



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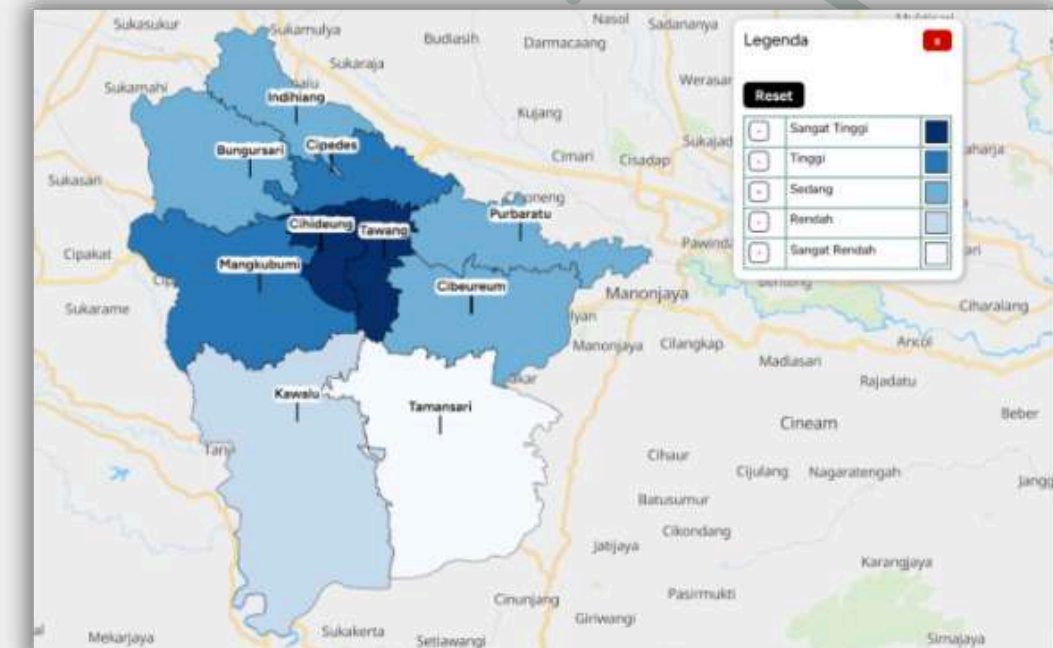
ACADEMY PROJECT

INTERNSHIP INDIVIDUAL PROJECT (2025)

Spatial Analysis for Mapping Potential Labor Absorption Areas by Sub-District using Criteria Importance Through Intercriteria Correlation (CRITIC) Method in Tasikmalaya, Jawa Barat, 2024

BACKGROUND OF RESEARCH

- Tasikmalaya's economy (10 sub-districts) is driven mainly by trade, construction, and manufacturing.
- A high participation rate (68.92%) contrasts sharply with job seekers (2,619) outnumbering available roles (1,067).
- Unequal job distribution across sub-districts worsens regional inequality and fuels unemployment.



Kelas	Range	Skor	Bobot	Skor x Bobot
Sangat Tinggi	387 - 394	5		0.289
Tinggi	344 - 386	4		0.176
Sedang	299 - 311	3	0.28	0.867
Rendah	211 - 281	2		0.226
Sangat Rendah	185 - 212	1		0.144

(a)

Kelas	Range	Skor	Bobot	Skor x Bobot
Sangat Tinggi	13.41 - 11.136	5		0.737
Tinggi	11.136 - 4.873	4		0.589
Sedang	4.873 - 2.354	3	0.18	0.442
Rendah	2.354 - 1.972	2		0.295
Sangat Rendah	1.972 - 1.868	1		0.147

(b)

Kelas	Range	Skor	Bobot	Skor x Bobot
Sangat Tinggi	9138 - 8324	5		0.855
Tinggi	8109 - 2941	4		0.642
Sedang	3945 - 2543	3	0.28	0.482
Rendah	2548 - 1736	2		0.321
Sangat Rendah	1708 - 1579	1		0.181

(c)

Kelas	Range	Skor	Bobot	Skor x Bobot
Sangat Tinggi	287 - 212	5		0.823
Tinggi	212 - 79	4		0.666
Sedang	79 - 43	3	0.17	0.51
Rendah	43 - 22	2		0.333
Sangat Rendah	22 - 21	1		0.167

(d)

Kelas	Range	Skor	Bobot	Skor x Bobot
Sangat Tinggi	0.73 - 0.88	5		0.609
Tinggi	0.88 - 0.53	4		0.487
Sedang	0.53 - 0.32	3	0.12	0.369
Rendah	0.32 - 0.21	2		0.244
Sangat Rendah	0.21 - 0.1	1		0.122

(e)

Kelas	Range	Skor	Bobot	Skor x Bobot
Sangat Tinggi	12324 - 28792	5		0.574
Tinggi	28792 - 25076	4		0.459
Sedang	25076 - 18392	3	0.21	0.344
Rendah	18392 - 12854	2		0.23
Sangat Rendah	12854 - 12858	1		0.115

(f)

Labor absorption in Tasikmalaya is concentrated in the **city center** due to **strong infrastructure**, while **peripheral areas lag behind**.

(a) Job Seekers; (b) Industrial Density; (c) Productive Age Pop;
(d) Job Vacancies; (e) Industrial Road Access; (f) Higher Education Pop

OPEN PROJECT

CERTIFICATIONS



linkedin.com/in/nuryabillautami



nuryabillau@gmail.com



082262904885



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THANK YOU

THANK YOU

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[linkedin.com/in/nuryabillautami](https://www.linkedin.com/in/nuryabillautami)



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nuryabillau@gmail.com



082262904885



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